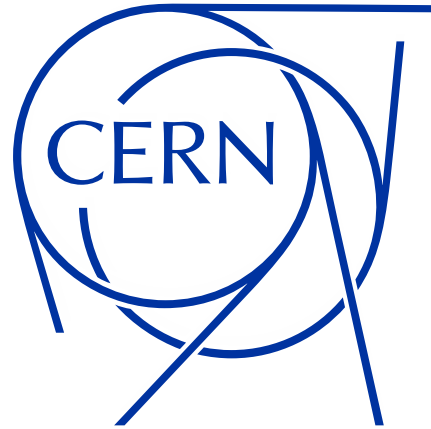
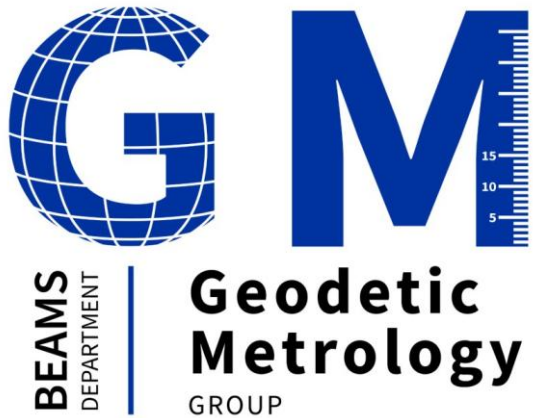




Alignment measurement at CERN : automatization of the ecartometry measurement using photogrammetry for the Large Hadron Collider

F. Barcet, P. Bestmann, D. Mergelkuhl CERN BE-GM

11/03/2025

Outline



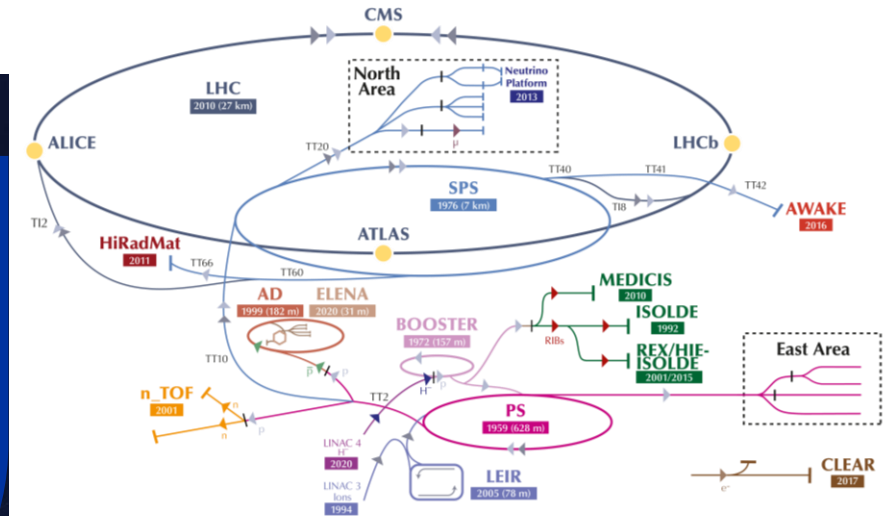
- **Introduction of CERN complex and constraints**
- **Survey train process**
 - Camera and inclinometer calibration
 - Acquisition and measurement
 - Calculation with MMVII and output
- **Results and analysis**
 - Comparison with Laser Tracker
 - Repeatability tests
- **Outlook and Conclusion**

Our mission



CERN is the world's biggest laboratory for particle physics.

Our goal is to understand the most fundamental particles and laws of the universe.



AD/ELENA: Antiproton Decelerator for antimatter studies

CAST, OSQAR: axions

CLOUD: impact of cosmic rays on clouds → implications on climate

COMPASS: hadron structure and spectroscopy

ISOLDE: radioactive nuclei facility

NA61/Shine: heavy ions and neutrino targets

NA62: rare kaon decays

NA63: interaction processes in strong EM fields in crystal targets

NA64: search for dark photons

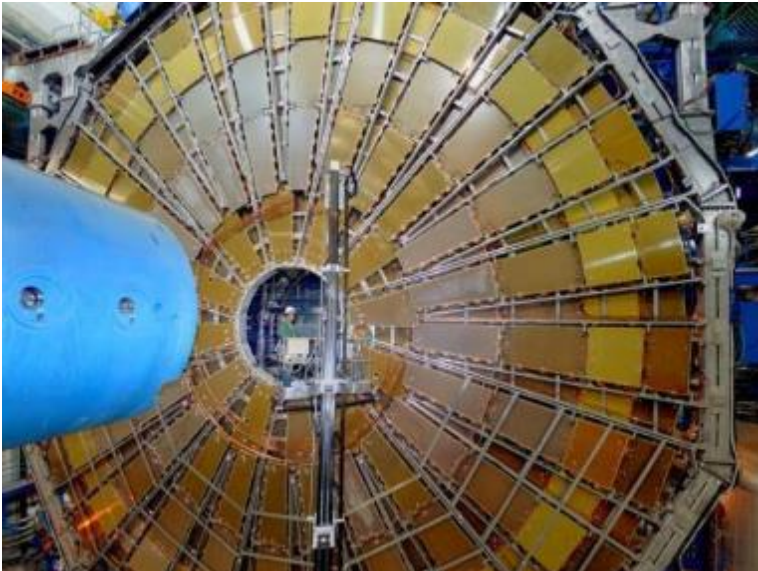
NA65: τ -neutrino production from D_s decays

Neutrino Platform: ν detectors R&D for experiments in US, Japan

n-TOF: n-induced cross-sections

UA9: crystal collimation

GM – Geodetic Metrology -- Our day to day challenges --



Size of the components



Severe environment...



Accuracy and precision

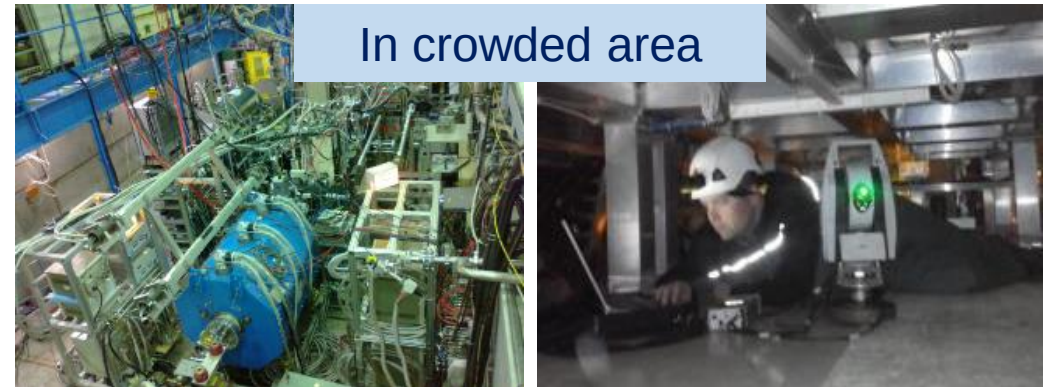
From μm to mm ...

Difficult configuration: straight and narrow



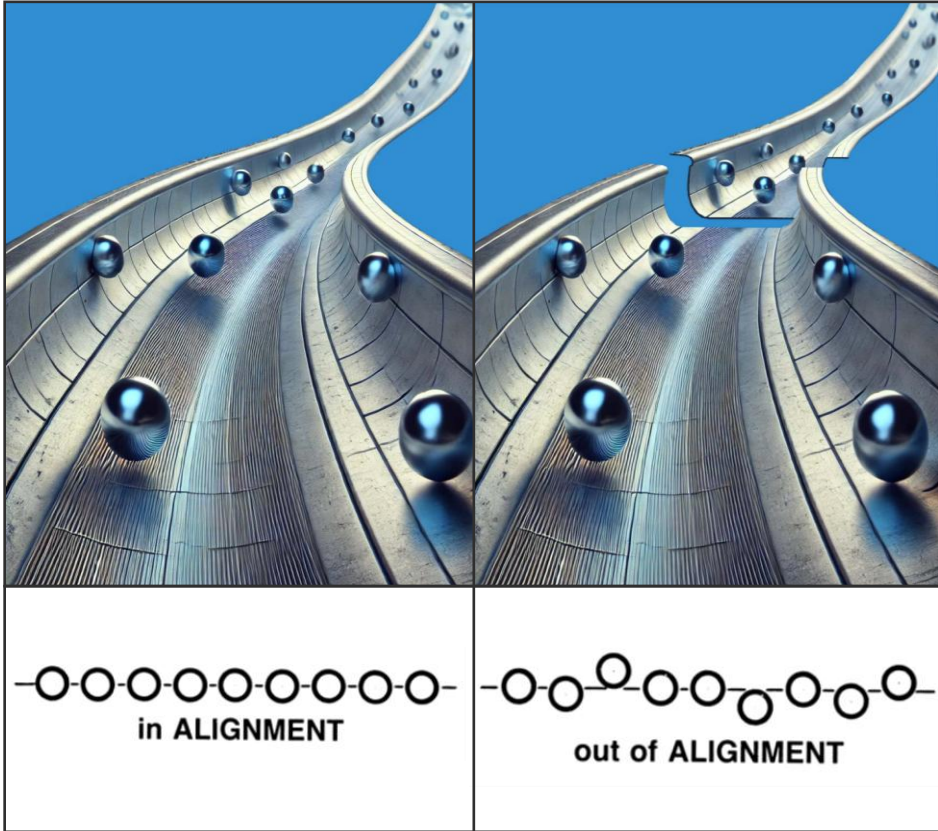
Underground

In crowded area



Why do we need high precision alignment?

To make the beam pass «through»: the components must be aligned in such a way that the beam is detected by beam position monitor, in order to launch «Beam based alignment»



The Earth is in constant motion:

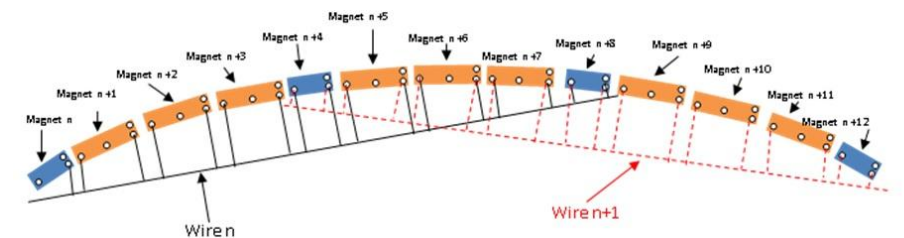
- ✓ Frequency of realignment as function of the alignment requirements and amplitude of ground motion
- ✓ Alignment tolerances: error of placement which, if exceeded, leads to a machine that is uncorrectable – with an unacceptable loss of luminosity

Solutions :

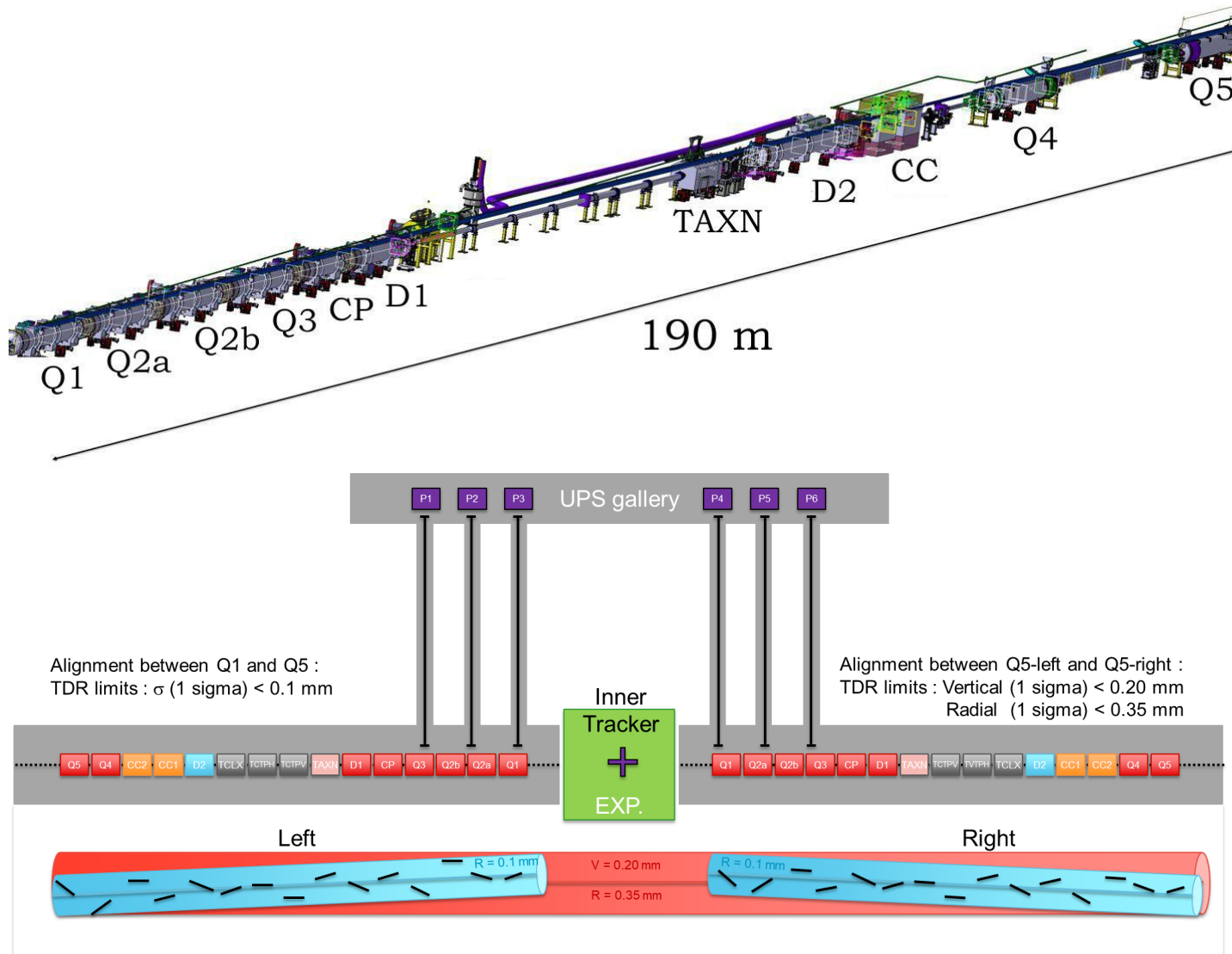
Optical & digital levels



Ecartometry



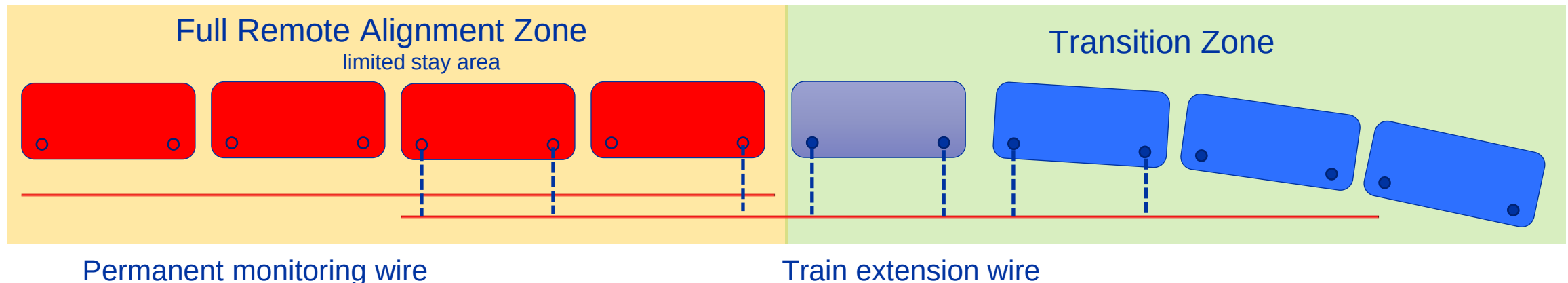
Upgrade to the High-Luminosity LHC under way



- The HL-LHC will use new technologies to provide 10 times more collisions than the LHC.
- It will start operating in 2030, and run until 2042.
- Alignment precision is required to be within 0.20 mm vertically and 0.35 mm radially over 420 meters.
- The ongoing upgrades present new challenges, particularly in maintaining alignment under increased radiation levels.
- A permanent system called the FRAS monitors this area.

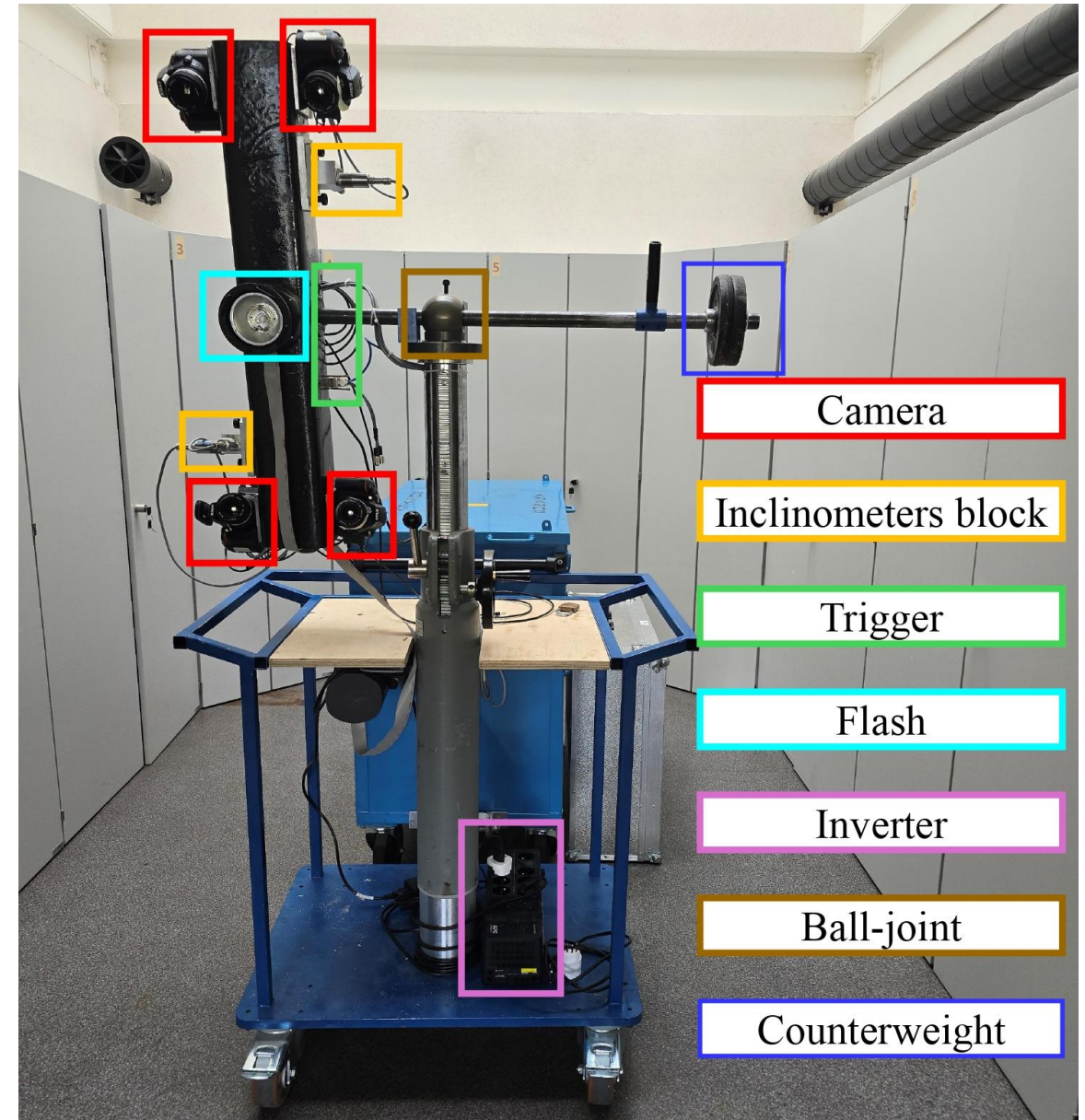
Limitations of Traditional Alignment Methods

- Wire measurements are necessary for accelerator alignment :
 - laser tracker : angular precision of $3\text{ }\mu\text{rad}$.
 - 120 m stretched wire : precision of $50\text{ }\mu\text{m}$ for an offset = angular accuracy of $0.8\text{ }\mu\text{rad}$
- Manual wire offset measurements are labor-intensive, time-consuming, and excluded due to high radiation levels.
- To maintain a continuous and smooth alignment, a link must be made between the LHC arcs and the FRAS.
- Given the increased radiation in these areas, traditional manual measurements are no longer feasible.

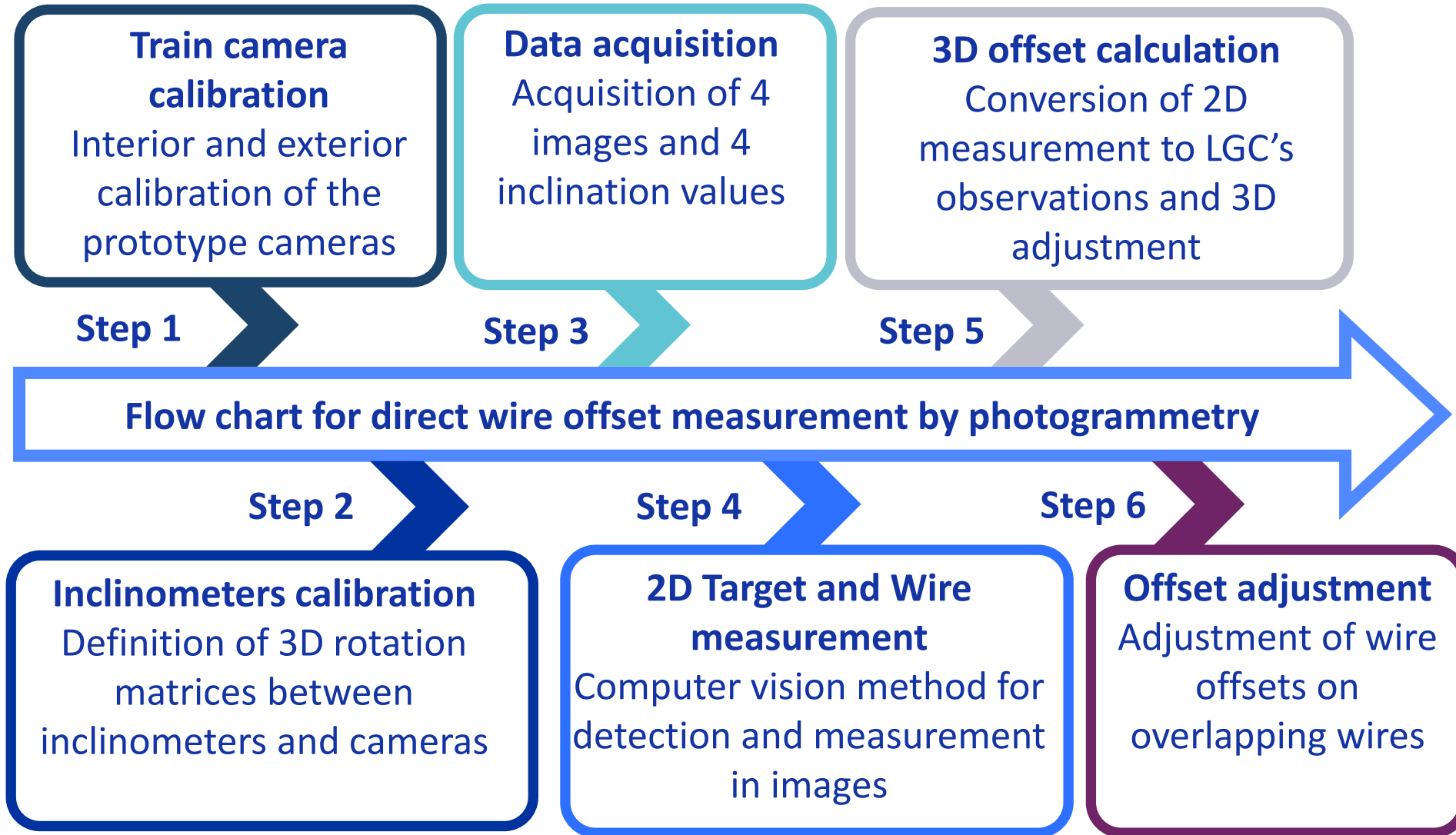


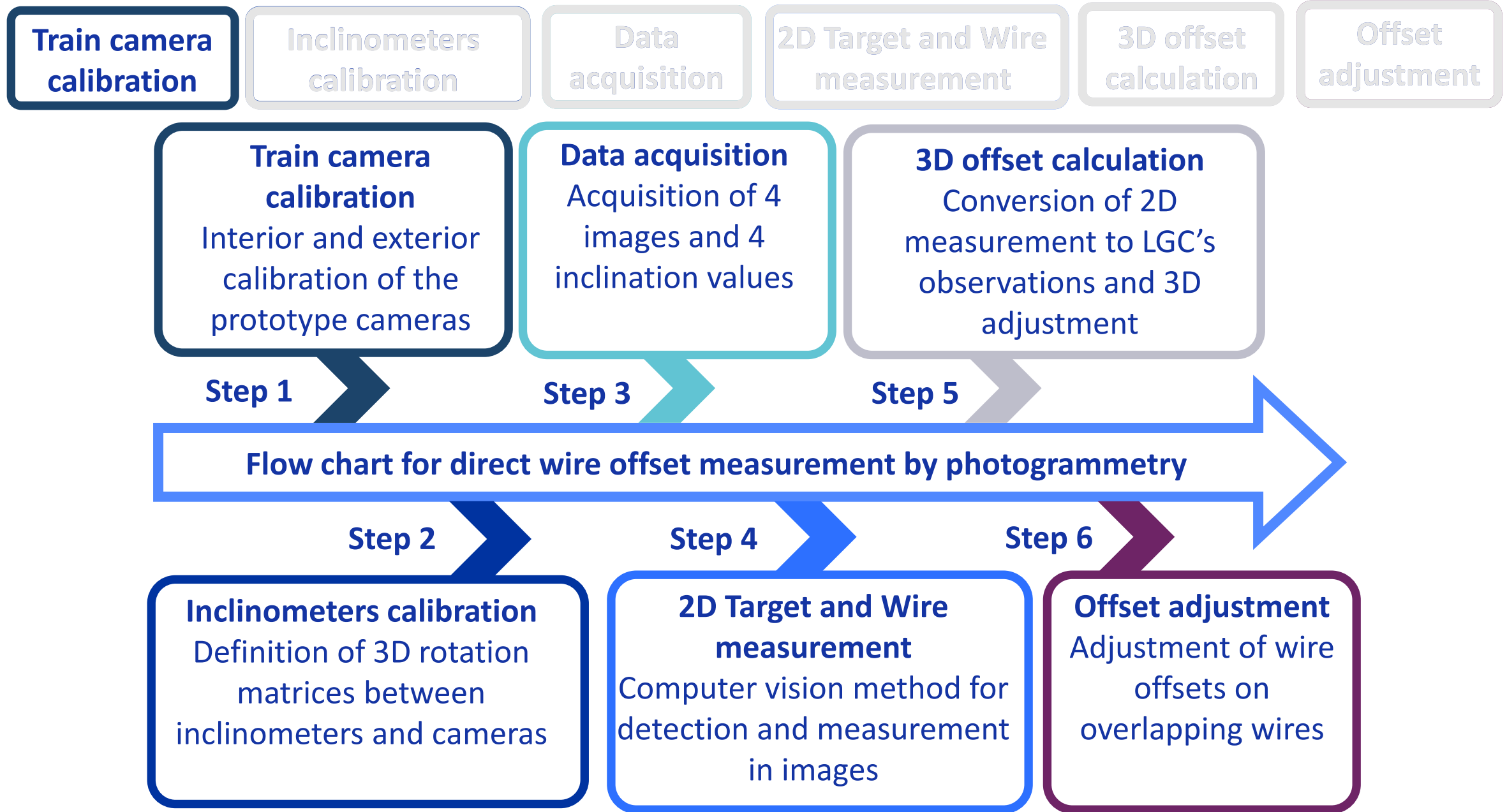
Hardware and Software Innovations

- Photogrammetry provides a non-contact method for high-precision measurements.
- The key to the process is the identification and measurement of targets and the stretched wire using advanced machine vision techniques.
- Integration with gravity through inclinometer calibration ensures accuracy.

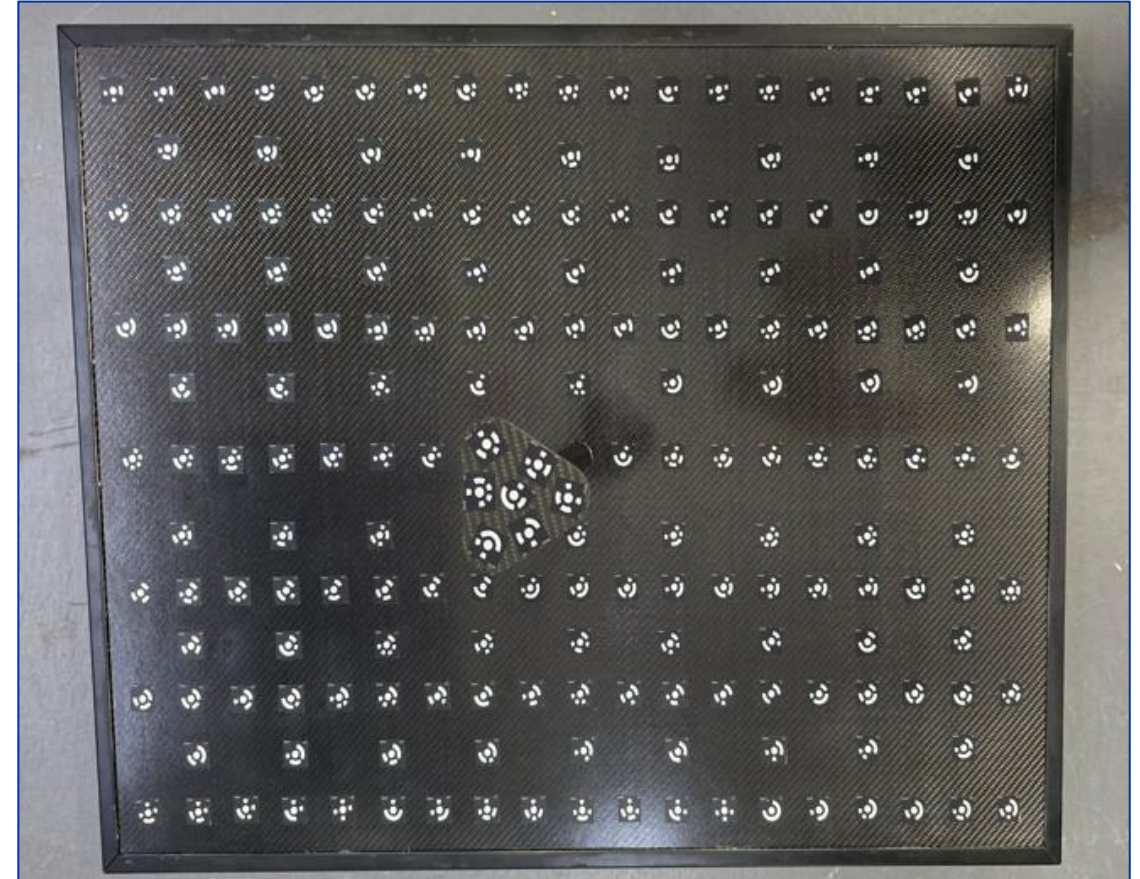
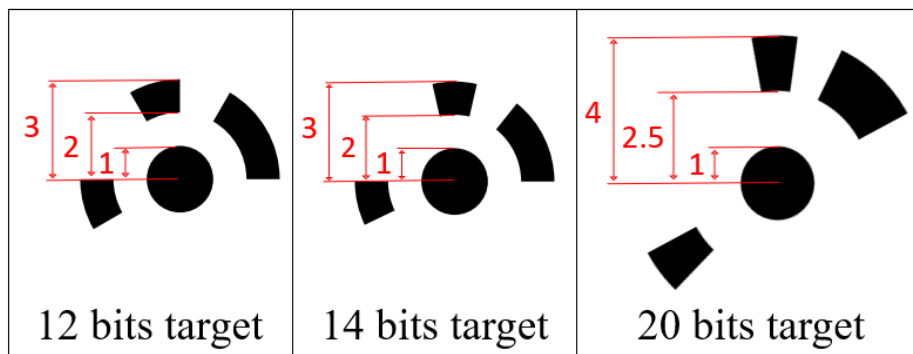


Process





- Calibration involves defining the interior and exterior orientation of the cameras. This can be performed by DPA-Studio (Hexagon) or MicMac (IGN).
- The carbon sandwich panel consists of a 700 mm × 500 mm with 200 coded targets glued on it.
- Regular calibration is essential for maintaining accuracy over time.



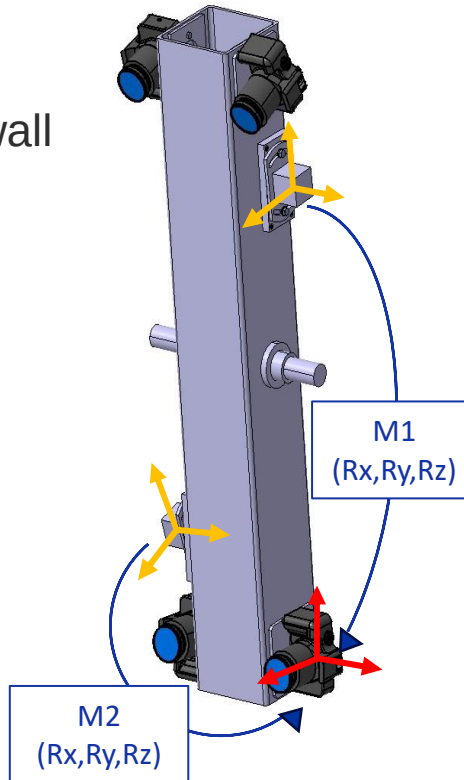
The inclinometers are calibrated to link the photogrammetric system to gravity, enabling precise vertical and horizontal measurements.

This procedure follows these steps:

- Installation of a dozen drift-nests for 1.5-inch SMRs on a wall
- Measurement by laser tracker
- The train setup positioned in front of the network
- Acquisition of 15 positions in multiple orientations
- Constraint adjustment in LGC
- Extraction of two rotation matrices



Wylrag.com



Train camera
calibration

Inclinometers
calibration

Data
acquisition

2D Target and Wire
measurement

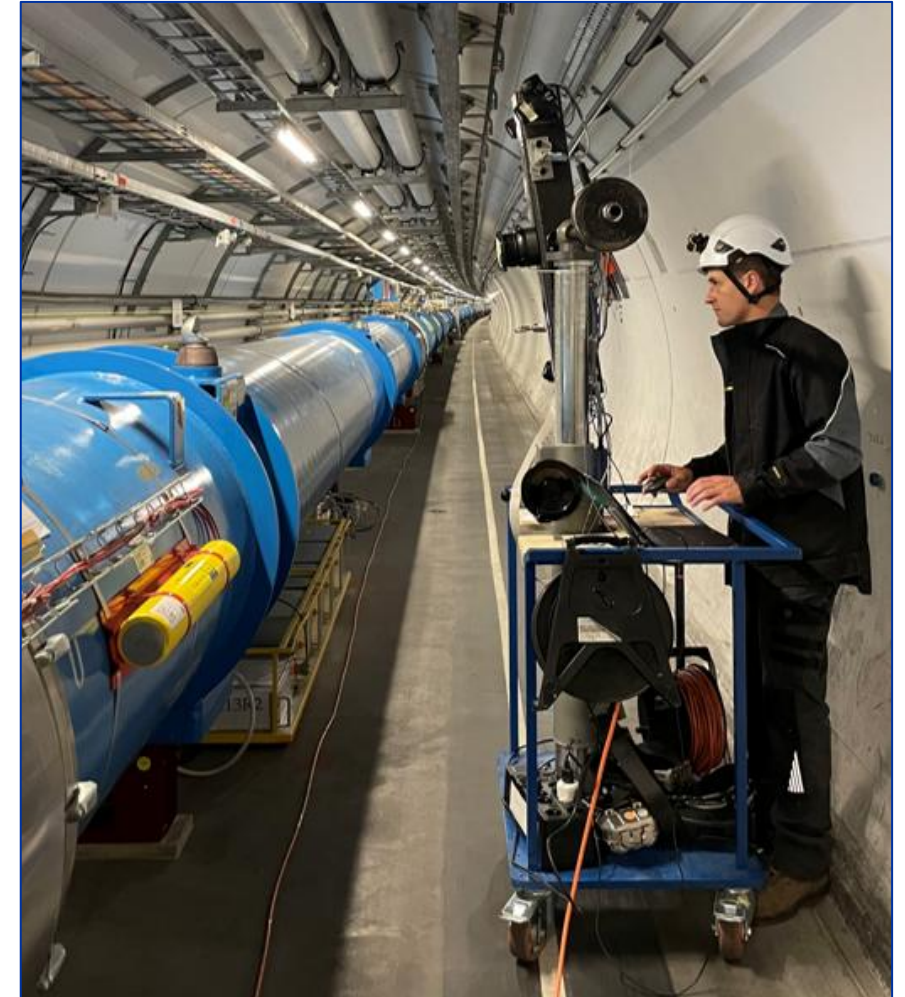
3D offset
calculation

Offset
adjustment

Wire stretching

Target installation on the fiducials

Multiple sets of 4 images taken in
front of each fiducials



Train camera
calibration

Inclinometers
calibration

Data
acquisition

2D Target and Wire
measurement

3D offset
calculation

Offset
adjustment

Target detection and measure

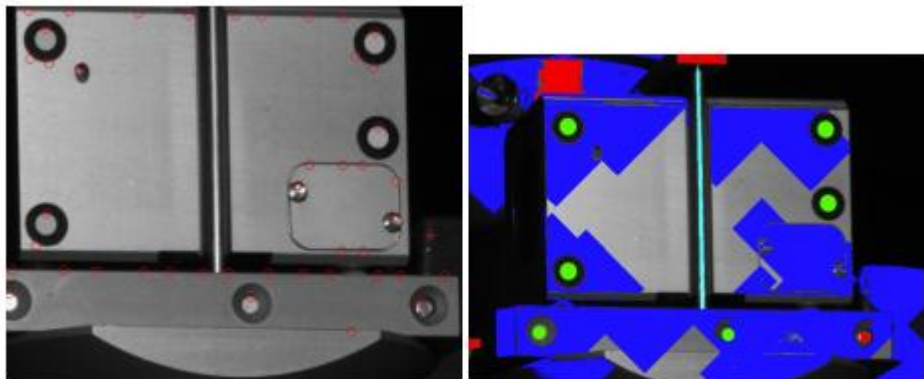


Figure 5.3: Seed points and connected component

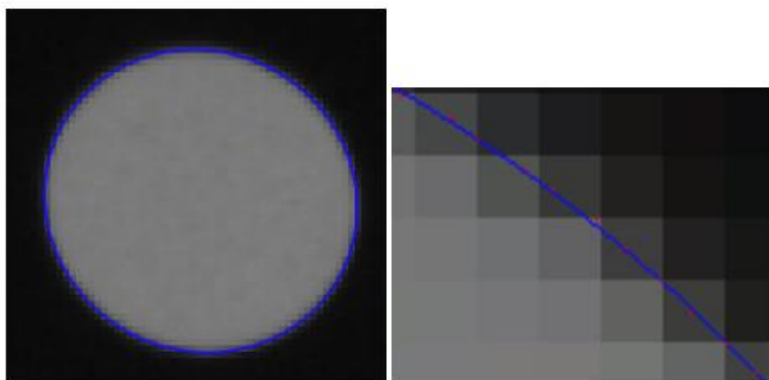


Figure 5.4: An extracted ellipse, and a zoom on the frontier : blue ellipse fitted on frontier points in red

Wire detection



Figure 10.1: An extract of image, the sobel gradient, non linear radiometric enhance of gradient

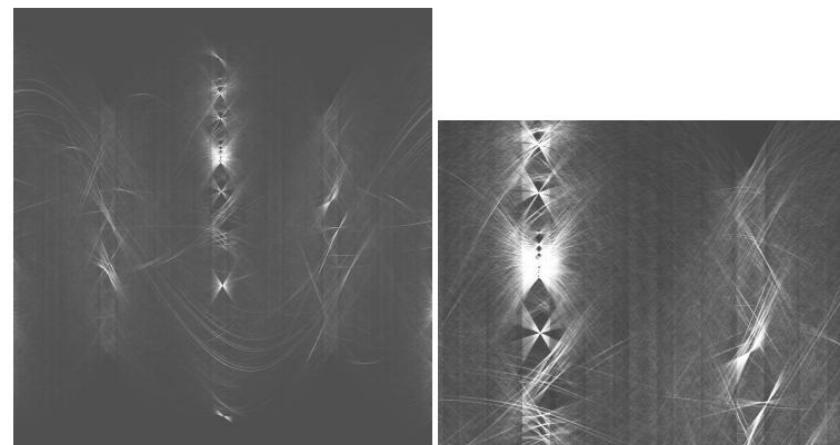
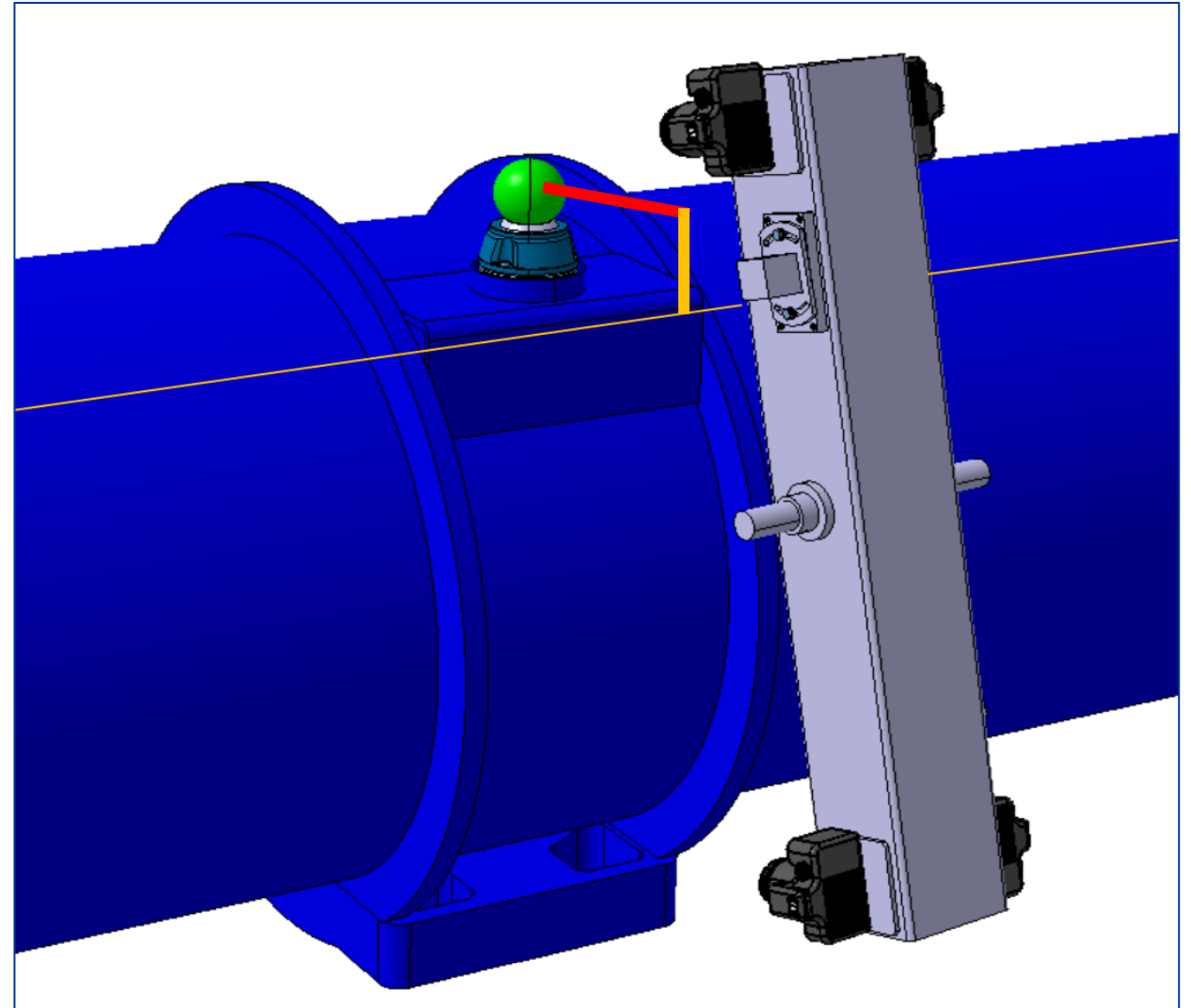


Figure 10.3: Example of hough transform, and a crop on it

Method from MMVII

- Bundle adjustment with MMVII with the rigid bloc and inclinometer constraints
- Extraction of the shortest horizontal distance between the wire and the target



Train camera
calibration

Inclinometers
calibration

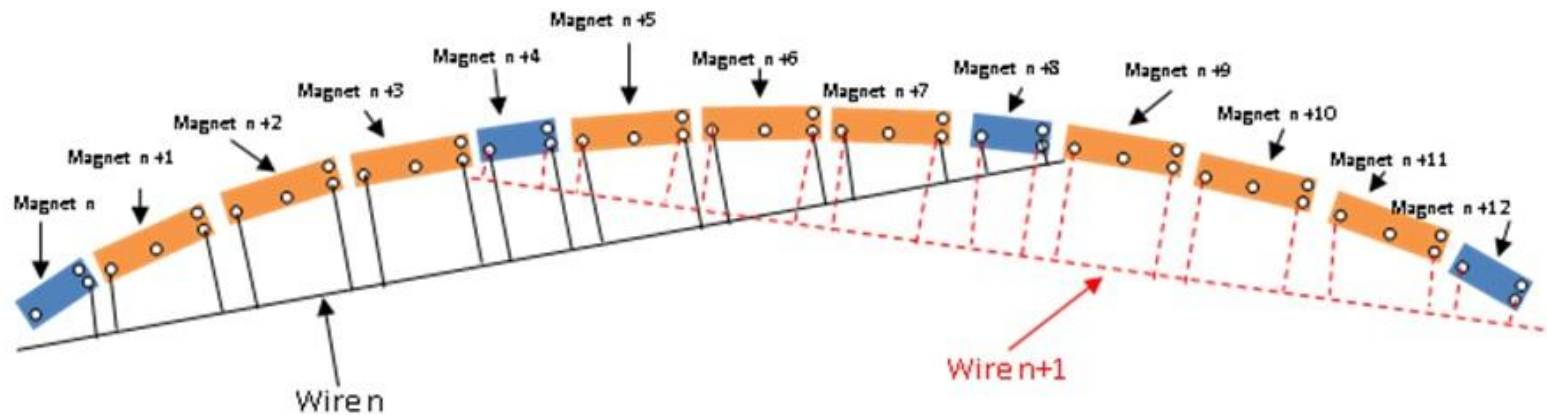
Data
acquisition

2D Target and Wire
measurement

3D offset
calculation

Offset
adjustment

- Extraction of individual offsets and associated precision
- Completion of the LGC file for wire offset calculation
- Least squared adjustment





LGC Report

LGC2 v2.8.beta.1_Apriori_covariances
LGC2 Copyright 2022, CERN SU. All rights reserved.

File name:
Line adjustment with survey train measurement

Source file name:
//eosproject-smb/eos/project/b/be-gm-survey-train/Data/Photo/TS1/Calcul/ECHO - LT
LGC/Echo_LT_TS1.lgc2

Computed on September 26, 2024 at 04:05:13 PM GMT+2 with processing time of 0.3663s.

Statistics

Number of observations: 155
Number of unknowns: 39
Number of constraints: 4
Number of equations: 155
Sigma zero a posteriori: 0.993086
Critical value: (0.8714, 1.1284)
Number of iterations: 3

Dataset

Number of frames: 1
Unknowns in frames introduced: 0
Number of points: 11
Unknowns in points introduced: 33
Number of angles: 2
Unknowns in angles introduced: 2
Number of planes: 2
Unknowns in planes introduced: 4

Point type numbers:

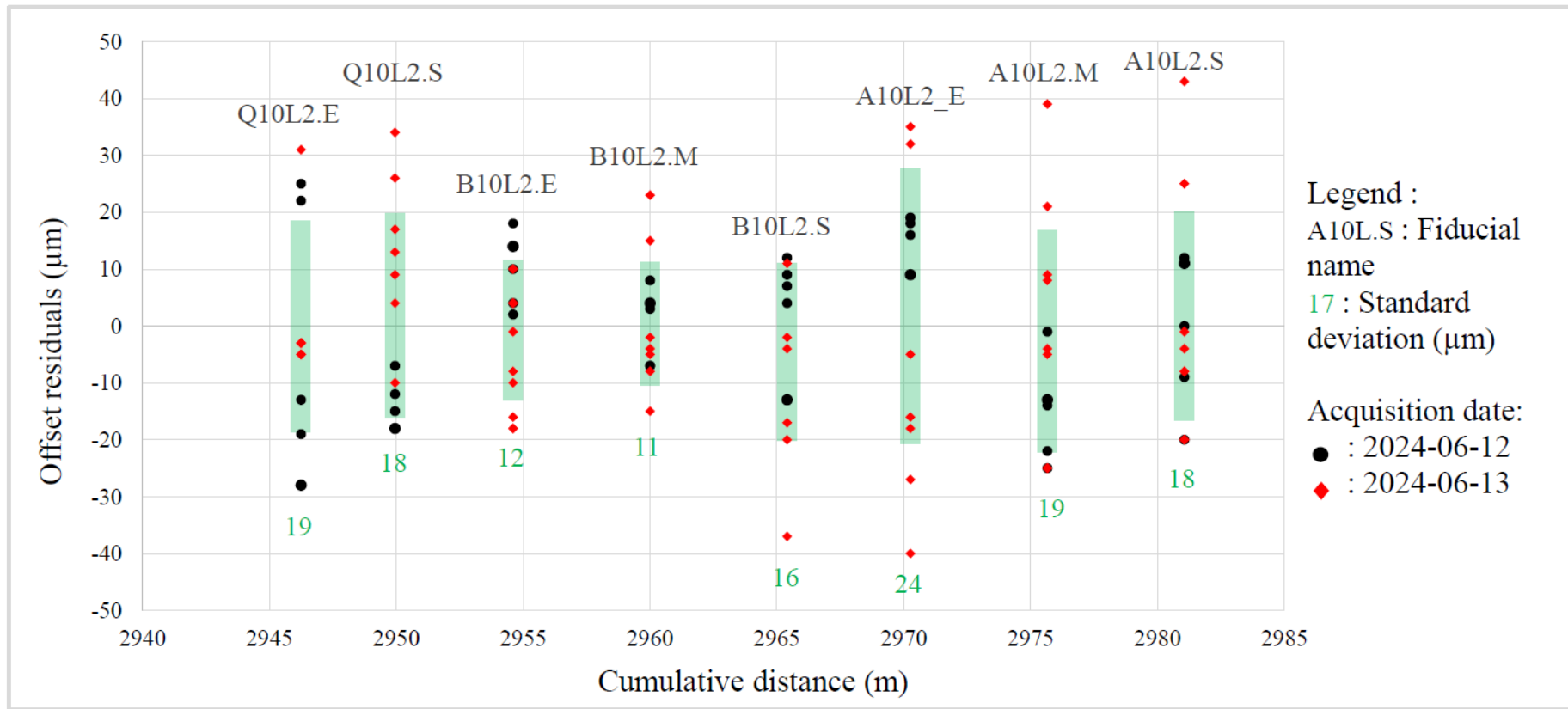
NumPoint: 11

Measurement type numbers:

NumANGL: 20
NumZEND: 20
NumDIST: 20
NumECHO: 95

Calculation with Laser Tracker and Train measurement

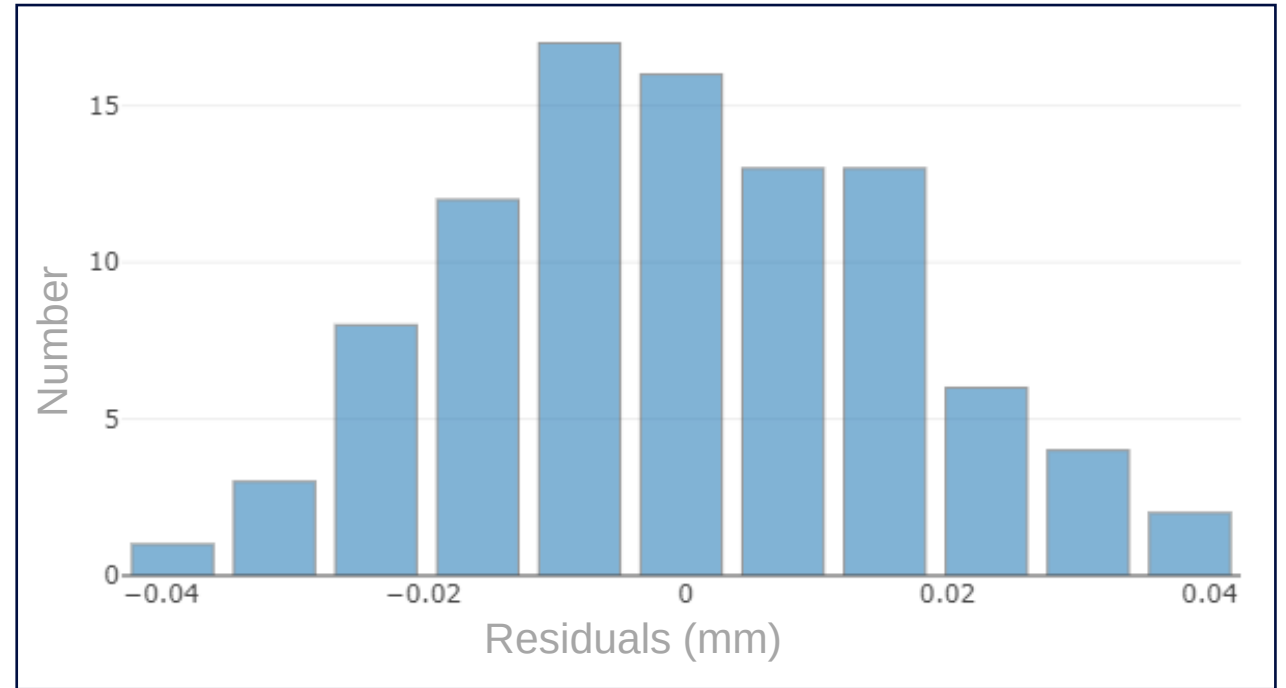
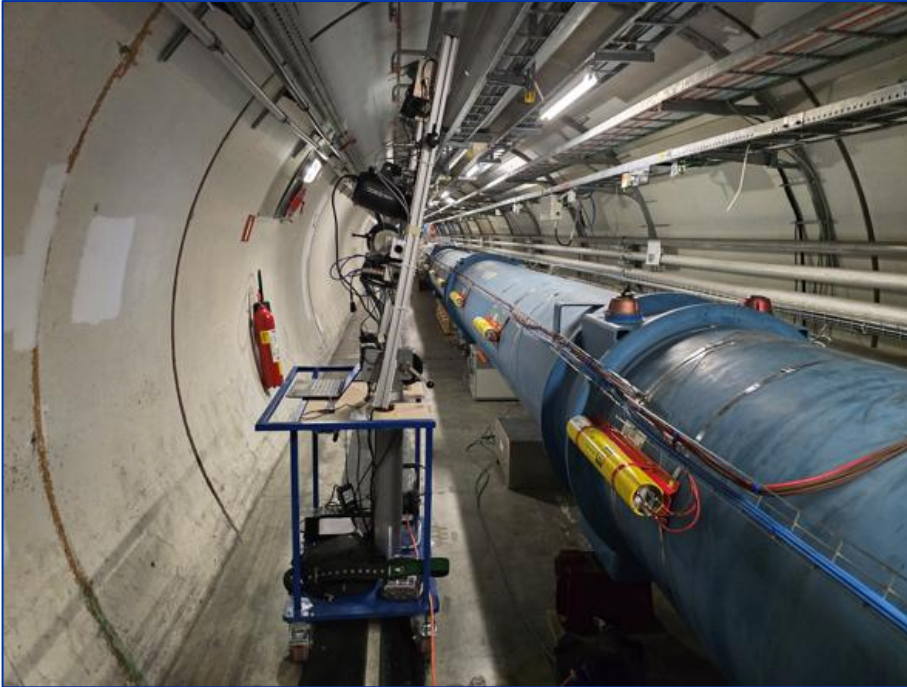
The standard deviation for 95 observation is $\sim 20 \mu\text{m}$. The specification of an automated system $< 50 \mu\text{m}$ for horizontal offset measurement seems reachable.



Early Repeatability Results

Tests were conducted in the LHC and SPS environments, with promising results.

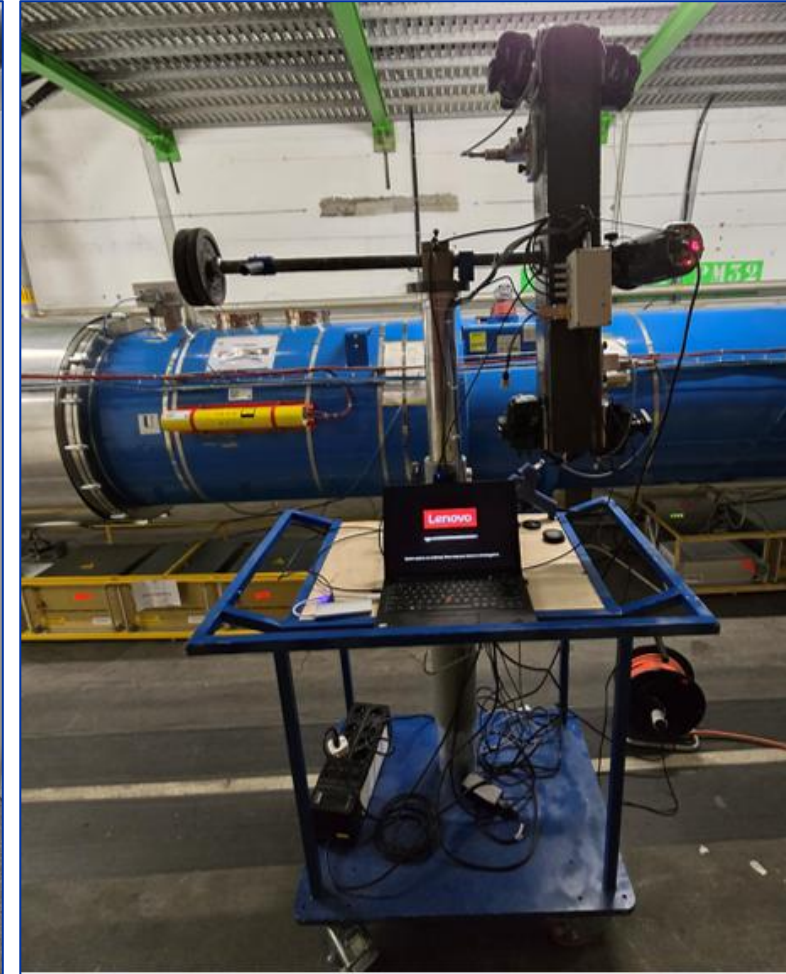
The initial tests have shown excellent repeatability, with a precision of 50 μm or better.



Histogram of residuals for 95 offsets observation with survey train

Development Timeline and Key Milestones

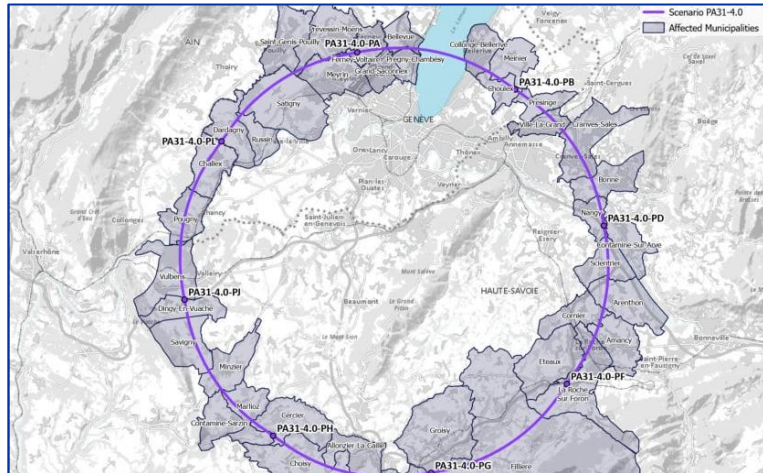
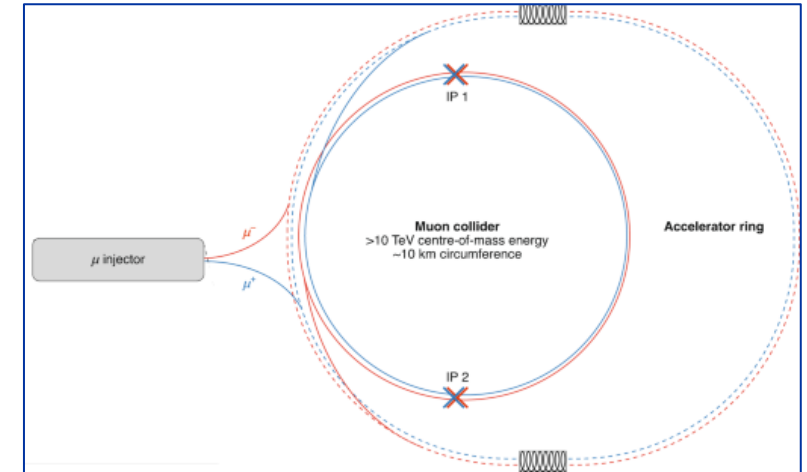
- The development is planned in stages, with final qualification tests scheduled for 2025 with the IT String test.
- The system must be ready in time for the different tasks during the Shutdown periods (LS3).
- Integration into TIM and full automation are the next major milestones.



Extending the System to Future Accelerators

The principles developed for HL-LHC alignment are applicable to other accelerators:

- SPS
- The arc of the LHC
- Muon Collider
- Future Circular Collider (FCC)



Conclusion

- Direct photogrammetry reduces collision risks and enhances measurement stability.
- The project is well on track, with promising future developments.
- The survey train and photogrammetry-based measurement system have proven effective in achieving the required precision.
- The system needs to stay modular and universal, even if specific tasks are targeted. It paves the way for fully automated measurement in high-radiation environments.



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